



Exam #1 Study Guide

15 True/False questions (1 point each) + 5 Numerical Free Response questions (5 points each)

75 Minutes, 40 Points

Key: Understand terminology & notation so you can excel at word problems.
ie. There are many ways to say predictors, *MSE*, error sum of squares, etc.

The listed topics are for numerical Qs. Conceptual Qs will show up from any section.

Statistical Learning (1 Q):

- Bayes Classification Error
- K-Nearest Neighbors

Simple Linear Regression (2 Qs):

- Least squares coefficient estimation and R^2
- Model predictions, calculator usage
- Residuals, MSE, SST, SSE, SSReg, R^2 calculations

Simple Linear Regression Inference (2 Qs):

- t -statistic, hypothesis tests, p -values
- Variance/standard error of coefficient estimates
- Confidence/Prediction Intervals

Vocabulary to Know:

- Response vs. Explanatory variable
- Regression vs. Classification
- Prediction vs. Inference
- Training vs. Test data
- Total Sum of Squares vs. Error Sum of Squares (aka. Residual Sum of Squares) vs. Regression Sum of Squares and relationship with Residual, Mean Squared Error, Residual Standard Error, R^2
- Mean Squared Error vs. Residual Standard Error
- Stochastic vs. Non-stochastic variable
- Heteroscedastic vs. Homoscedastic
- Null vs. Alternate Hypothesis
- Variance vs. Standard Error
- p -value vs. critical α vs. t -statistic and relationship w/hypothesis testing and conclusion
- Confidence vs. Prediction interval



Formulas to Remember:

- Bayes Error Rate
- Distance Formula for KNN
- $b_0 = \bar{y} - b_1 \bar{x}$
- $b_1 = r \frac{s_y}{s_x} = \frac{\text{Cov}(X,Y)}{s_x^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$ (will be provided if needed)
- Residual: $e_i = y_i - \hat{y}_i$
- $SST = SSE + SSReg$
- $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- $SST = \sum_{i=1}^n (y_i - \bar{y})^2$
- $SSReg = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = b_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 = b_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- $MSE = s^2 = \frac{SSE}{n-2}$
- $s_y^2 = \frac{SST}{n-1}$
- $r = \frac{\text{Cov}(X,Y)}{s_x s_y}$ and using your calculator
- $R^2 = \frac{SSReg}{SST}$
- $se(b_0) = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- $t = \frac{\text{Estimator} - H_0 \text{ value}}{se(\text{Estimator})}$
- Confidence Interval: $b_1 \pm t_{n-2, \frac{\alpha}{2}} * se(b_1)$
- $se(\hat{y}_{pred}) = s \sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- Prediction Interval: $\hat{y}_{pred} \pm t_{n-2, \frac{\alpha}{2}} * se(\hat{y}_{pred})$